Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [12CS/IS-832]

EIGHTH SEMESTER B.E EXAMINATION, JUNE – 2016

STORAGE AREA NETWORKS

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1.	a.	Explain the logical components and physical components of connectivity in a data center.	07
	b.	Explain the evolution of a storage technology and architecture.	07
2.	a.	Define seek time. Explain what are the other specifications of a seek time in a hard disk.	05
	b.	Explain the disk native command queuing and the algorithms being used with suitable examples.	05
	c.	The average Input/Output size of an application is 32KB. The following specifications are available from the disk manufacturer average seek time=5ms in a random I/O environment, 15,000rpm, transfer rate is 40MB/s. Determine the maximum IOPS that could be performed with this data for this application. Taking this case as an examples explain the relationship between disk utilization and IOPS.	04
3.	a.	With a neat diagram explain RAID4. How a write penalty occurs on modification of a single data block? Explain.	08
	b.	How does a hotspare disk aid when one of the disk fails? Explain all the phases briefly.	06
4.	a.	With neat diagrams, explain how the frames are transmitted in class 2 and class 3 services.	07
	b.	Explain the Fibre-channel stack-layer 2 of fibre channel frame format with a neat diagram.	07
5.	a.	Differentiate integrated NAS and gateway NAS connectivity implementations with neat diagrams.	07
	b.	How is data transfer occur in a NAS? Explain all the NAS components with a neat diagram.	07
6.	a.	Brief about the components of a CAS architecture.	06
	b.	Define replication process in the storage Area Network	03

c. List the differences between backup and archive. 05

7. a. Define Information Availability IA. How is information availability measured? 05

b. A department requires to access a database application from Monday to Friday 9AM-6PM. Last Friday at 10AM the application has crashed and it took 8 hours to fix the problem. What is the availability of the application during a week? 04

c. What is a single point of failure in storage networks? 05

8. a. Explain the various forms of storage management. 05

b. What is storage tiering? Explain the different forms of storage tiering. 05

c. What is monitoring in storage management? 04